

Do individual stocks respond to volatility shocks beyond the market move?

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Summary

I test whether the daily returns of nine large US stocks respond to changes in the VIX once the market return is already accounted for. I expected a negative response. Over about twenty years of daily data I do not find one. The estimated volatility exposures are small, they are split between negative (four stocks) and positive (five stocks), and none is statistically different from zero. The market return explains nearly everything the VIX change appeared to explain.

Question

A rise in the VIX usually comes on a day when the S&P 500 falls, so a stock that drops on a high VIX day might only be reacting to the market. The question is whether there is anything left over: does a stock move with the part of the VIX change that the market return does not explain? My hypothesis going in was that this leftover exposure would be negative for most stocks.

Data

Daily data for the S&P 500 index, the VIX, and nine stocks (AAPL, MSFT, GOOGL, NVDA, JPM, XOM, JNJ, KO, WMT) were pulled through the Interactive Brokers API. I picked the nine to cover both cyclical and defensive names. After aligning all eleven series to common dates, the panel has 5,021 trading days from 4 October 2006 to 25 September 2026. The volatility variable is the daily change in the VIX close, in VIX points. I use the change and not the level because the level is highly persistent, and regressing daily returns on a persistent series can produce results that are not real.

Method

The daily VIX change and the S&P 500 return have a correlation of -0.82 in this sample, so putting both into one regression makes the two coefficients hard to read separately. I deal with this in two steps.

Step 1. Regress the VIX change on the S&P 500 return and keep the residual:

$$dVIX(t) = a + b * SPX(t) + u(t)$$

The slope is -131, meaning a 1 percent fall in the index goes with a rise of about 1.3 VIX points, and the R-squared is 0.67. The residual $u(t)$ is the part of the VIX change not explained by the market that day. It is uncorrelated with the S&P 500 return by construction and has a standard deviation of 1.12 points, against 1.96 for the raw change.

Step 2. For each stock, regress its return on the market return and the residual from step 1:

$$r(i,t) = \alpha(i) + \beta_{mkt}(i) * SPX(t) + \beta_{vol}(i) * u(t) + e(i,t)$$

β_{vol} is the number of interest: the change in the stock return for a one point volatility shock, holding the market return fixed. Daily return regressions have residuals whose variance changes over time and which can be correlated across days, so ordinary standard errors are too small. I use Newey-West (HAC) standard errors with five lags. This does not change the coefficients, only the uncertainty around them.

Results

Stock	Market beta	Vol beta	OLS s.e.	HAC s.e.	t (HAC)	R-squared
NVDA	1.57	-0.00056	0.00030	0.00041	-1.37	0.39
XOM	0.87	-0.00036	0.00017	0.00028	-1.29	0.39
WMT	0.51	-0.00024	0.00015	0.00023	-1.05	0.22
KO	0.54	-0.00017	0.00012	0.00026	-0.63	0.31
MSFT	1.05	+0.00008	0.00015	0.00022	+0.38	0.53
JNJ	0.50	+0.00009	0.00012	0.00025	+0.37	0.31
GOOGL	1.02	+0.00030	0.00018	0.00027	+1.11	0.43
AAPL	1.06	+0.00033	0.00019	0.00033	+0.98	0.44
JPM	1.37	+0.00064	0.00020	0.00039	+1.63	0.53

Table 1. Step 2 estimates, 5,021 daily observations per stock, sorted by vol beta. Vol beta is the return per one point orthogonalized VIX change.

Four of the nine volatility betas are negative and five are positive. None has a t-statistic beyond 2 in either direction; the largest in absolute value is JPM at +1.63. The sizes are small as well. The most negative estimate, NVDA at -0.00056, means a one point volatility shock goes with a return about 0.06 percentage points lower. So the hypothesis is not supported: once the market return is in the regression, the leftover VIX change tells me little about these stocks on a typical day.

The market betas behave as expected, which is a check that the panel is built correctly. NVDA (1.57) and JPM (1.37) are the most sensitive to the market and JNJ, WMT and KO sit near 0.5.

The correction to the standard errors matters. The HAC standard errors are between about 1.3 and 2.2 times the ordinary ones. With ordinary standard errors JPM ($t = 3.2$), XOM ($t = -2.2$) and NVDA ($t = -1.9$) would look significant or close to it, and I would have reported an effect that is not there.

Stability over time

A full-sample coefficient near zero could hide an exposure that is negative in some years and positive in others. To check, I re-estimated step 2 on a rolling window of 252 trading days.

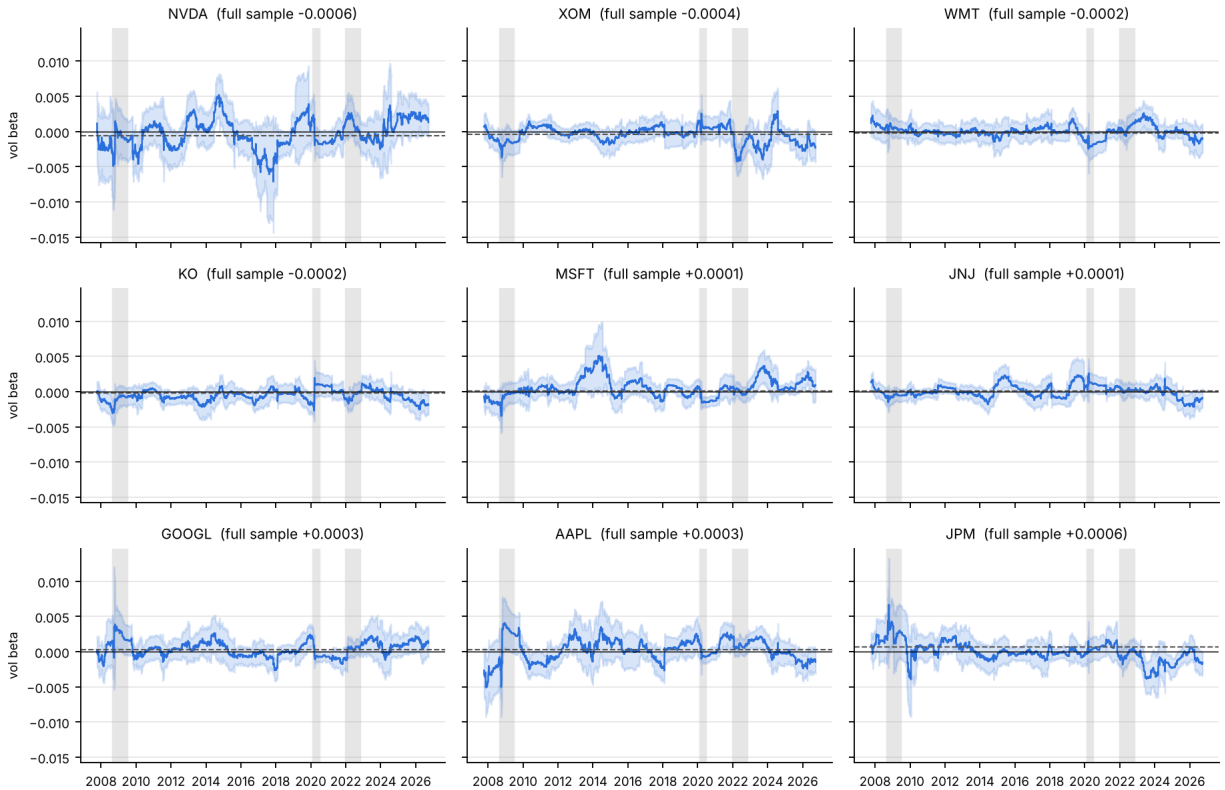


Figure 1. Rolling 252-day vol beta for each stock with a band of two heteroskedasticity-robust standard errors. Dashed line: full-sample estimate. Shaded: September 2008 to June 2009, February to June 2020, January to October 2022.

The rolling estimates do move, and they change sign within the same stock. NVDA is the widest, ranging from about -0.007 to +0.005. For most stocks the band covers zero for most of the sample. I do not see a consistent pattern in the three shaded stress periods: the estimates do not turn negative across the board in any of them.

Limitations

Nine stocks is a small and hand-picked group, all large and all still listed today, so these results say nothing about smaller firms or firms that failed. The step 1 regression uses the full sample, so the residual on any day uses information from later dates; that is fine for describing the data but would not be for a trading test. The relation between the VIX change and the market return is also not really a straight line, since the VIX reacts more to falls than to rises, and a linear step 1 leaves some of that in the residual. The five lag choice for the HAC errors is a common default that I did not test against other choices. The rolling windows overlap heavily, so neighbouring points in Figure 1 are not independent.

Next steps

The next things I would do are allow step 1 to differ for up and down days, repeat step 2 with other lag lengths, and extend the group of stocks well beyond nine so that the cross-section of volatility betas can be compared against firm characteristics such as size and sector.